

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 10 — Scalar Functions — String Operations
Customer & Product Database

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Roll No	2024-SE-11
Semester	04
Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	10
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Clean and standardize text data using TRIM, UPPER, and LOWER.
- Build dynamic strings using CONCAT.
- Extract substrings using SUBSTRING_INDEX, LEFT, and SUBSTRING.
- Search within strings using LOCATE, and transform them using REPLACE and LPAD.

2. Database Structure

The database (scalar_lab) uses Customer and Product tables containing intentionally messy data (inconsistent casing, extra whitespace) to practice string cleaning.

2.1 Tables

Table	Purpose
Customer	Customer name, email, city, phone, join date, and date of birth
Product	Product name, category, price, stock quantity, and launch date

3. Key Tasks / Concepts Implemented

- Task A1-A3: Removing extra whitespace with TRIM, standardizing case with UPPER/LOWER, measuring cleaned length with CHAR_LENGTH.
- Task A4: Building a personalized greeting message with CONCAT.
- Task A5-A6: Extracting the username and domain from an email using SUBSTRING_INDEX.
- Task A7-A8: Shortening names with LEFT, and masking phone numbers for privacy.
- Task A9-A10: Generating URL-friendly slugs with REPLACE, and zero-padding IDs with LPAD.
- Task A11-A12: Finding a substring's position with LOCATE, and splitting out a first name dynamically.

4. Implementation (phpMyAdmin)

All string function tasks were implemented and executed using phpMyAdmin under XAMPP.

Server: 127.0.0.1 » Database: scalar_lab » Table: customer

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Extra options

				CustID	CustName	Email	City	Phone	JoinDate	DOB
☐				1	Ali Khan	ali.khan@MAIL.com	Lahore	0300-1112233	2022-01-15	1995-04-12
☐				2	Sara Iqbal	sara@example.com	Karachi	0301-4445566	2022-04-22	1998-11-20
☐				3	HAMZA RAZA	hamza@example.com	Lahore	0302-7778899	2023-02-10	1997-08-05
☐				4	Ayesha Noor	NULL	Islamabad	0303-1234567	2023-05-18	1999-02-14
☐				5	bilal ahmed	bilal@MAIL.COM	Karachi	0304-2345678	2023-09-01	2000-06-30
☐				6	Fatima Sheikh	fatima@example.com	NULL	0305-3456789	2024-01-12	1996-10-25
☐				7	Usman Tariq	usman@example.com	Lahore	NULL	2024-06-30	2001-03-18
☐				8	Maira Javed	maira@example.com	Islamabad	0307-5678901	2024-08-25	1994-12-09

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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	CustID	int(11)		No	None			Change Drop More
☐	2	CustName	varchar(60) utf8mb4_general_ci		No	None			Change Drop More
☐	3	Email	varchar(80) utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4	City	varchar(30) utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5	Phone	varchar(20) utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6	JoinDate	date		Yes	NULL			Change Drop More
☐	7	DOB	date		Yes	NULL			Change Drop More

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Add to central columns Remove from central columns

Server: 127.0.0.1 » Database: scalar_lab » Table: product

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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	ProdID	int(11)		No	None			Change Drop More
☐	2	ProdName	varchar(60) utf8mb4_general_ci		No	None			Change Drop More
☐	3	Category	varchar(30) utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4	Price	decimal(10,2)		Yes	NULL			Change Drop More
☐	5	StockQty	int(11)		Yes	NULL			Change Drop More
☐	6	LaunchDate	date		Yes	NULL			Change Drop More

☐ Check all | With selected: Browse Change Drop Primary Unique Index Spatial Fulltext

Add to central columns Remove from central columns

Server: 127.0.0.1 » Database: scalar_lab » Table: product

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Extra options

			ProdID	ProdName	Category	Price	StockQty	LaunchDate
☐				101	Laptop Pro 15	Electronics	185000.00	12 2023-03-10
☐				102	Wireless Mouse	Electronics	2500.00	50 2022-07-22
☐				103	USB-C Cable	Electronics	800.00	100 2021-11-05
☐				104	Office Chair	Furniture	18500.00	8 2023-01-15
☐				105	Standing Desk	Furniture	45000.50	5 2024-02-28
☐				106	Notebook A4	Stationery	350.00	200 2020-04-01
☐				107	Ballpoint Pen 10pk	Stationery	450.00	150 2020-04-01
☐				108	Coffee Beans 1kg	Grocery	1899.99	30 2023-09-20
☐				109	Green Tea Box	Grocery	650.00	45 2022-12-12
☐				110	Bluetooth Speaker	Electronics	7500.00	18 2024-05-18

Structure and browser tabs for 2 tables

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Showing rows 0 - 7 (8 total, Query took 0.0003 seconds.)

```
-- Task A1: Stripping out unintentional leading and trailing whitespaces from names.
SELECT CustID, CustName, TRIM(CustName) AS CleanedName FROM Customer;
```

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Extra options

				CustID	CustName	CleanedName
☐				1	Ali Khan	Ali Khan
☐				2	Sara Iqbal	Sara Iqbal
☐				3	HAMZA RAZA	HAMZA RAZA
☐				4	Ayesha Noor	Ayesha Noor
☐				5	bilal ahmed	bilal ahmed
☐				6	Fatima Sheikh	Fatima Sheikh
☐				7	Usman Tariq	Usman Tariq
☐				8	Maira Javed	Maira Javed

TRIM

Server: 127.0.0.1 » Database: scalar_lab » Table: Customer

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```
SELECT CustID, UPPER(CustName) AS UpperName, LOWER(CustName) AS LowerName FROM Customer;
```

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☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	CustID	UpperName	LowerName
☐ Edit Copy Delete	1	ALI KHAN	ali khan
☐ Edit Copy Delete	2	SARA IQBAL	sara iqbal
☐ Edit Copy Delete	3	HAMZA RAZA	hamza raza
☐ Edit Copy Delete	4	AYESHA NOOR	ayesha noor
☐ Edit Copy Delete	5	BILAL AHMED	bilal ahmed
☐ Edit Copy Delete	6	FATIMA SHEIKH	fatima sheikh
☐ Edit Copy Delete	7	USMAN TARIQ	usman tariq
☐ Edit Copy Delete	8	MAIRA JAVED	maira javed

UPPER/LOWER

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```
SELECT TRIM(CustName) AS CleanedName, CHAR_LENGTH(TRIM(CustName)) AS NameLength FROM Customer;
```

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Extra options

CleanedName	NameLength
Ali Khan	8
Sara Iqbal	10
HAMZA RAZA	10
Ayesha Noor	11
bilal ahmed	11
Fatima Sheikh	13
Usman Tariq	11
Maira Javed	11

TRIM

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```
SELECT CustName, SUBSTRING_INDEX(Email, '@', 1) AS Username FROM Customer WHERE Email IS NOT NULL;
```

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Extra options

				CustName	Username
☐				Ali Khan	ali.khan
☐				Sara Iqbal	sara
☐				HAMZA RAZA	hamza
☐				bilal ahmed	bilal
☐				Fatima Sheikh	fatima
☐				Usman Tariq	usman
☐				Maira Javed	maira

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EMAIL Splitting query

```
SELECT CONCAT(LEFT(Phone, 4), '-XXX-XXXX') AS MaskedPhone FROM Customer WHERE Phone IS NOT NULL;
```

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Extra options

MaskedPhone

0300-XXX-XXXX
0301-XXX-XXXX
0302-XXX-XXXX
0303-XXX-XXXX
0304-XXX-XXXX
0305-XXX-XXXX
0307-XXX-XXXX

PHONE masking query

5. Conclusion

This lab built practical skill in cleaning and reshaping messy real-world text data directly in SQL, a technique used constantly in real reporting pipelines.